

WASP-36b

R-0993

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_180212 · created 2026-07-29T23:58:29+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458161.8318 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.048
Transit depth (Rp/Rs)^2	2.75 +/- 1.59 [%]
Orbital Inclination (inc)	84.94 +/- 2.9
Airmass coefficient 1 (a1)	2.16 +/- 1.21
Airmass coefficient 2 (a2)	-0.57 +/- 0.37
Scatter in the residuals of the lightcurve fit is	52.9 %
Best Comparison Star	None
Optimal Aperture	2.53
Optimal Annulus	5.0
Transit Duration (day)	0.083 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.048 vs 0.1368 (deviation 0.61σ)
Duration fit vs published	0.083 d vs 0.0758 d (deviation 0.38σ)
Mid-time O–C	-6.5 min
Depth SNR	3.5
Residual scatter	52.9 %

Light curve

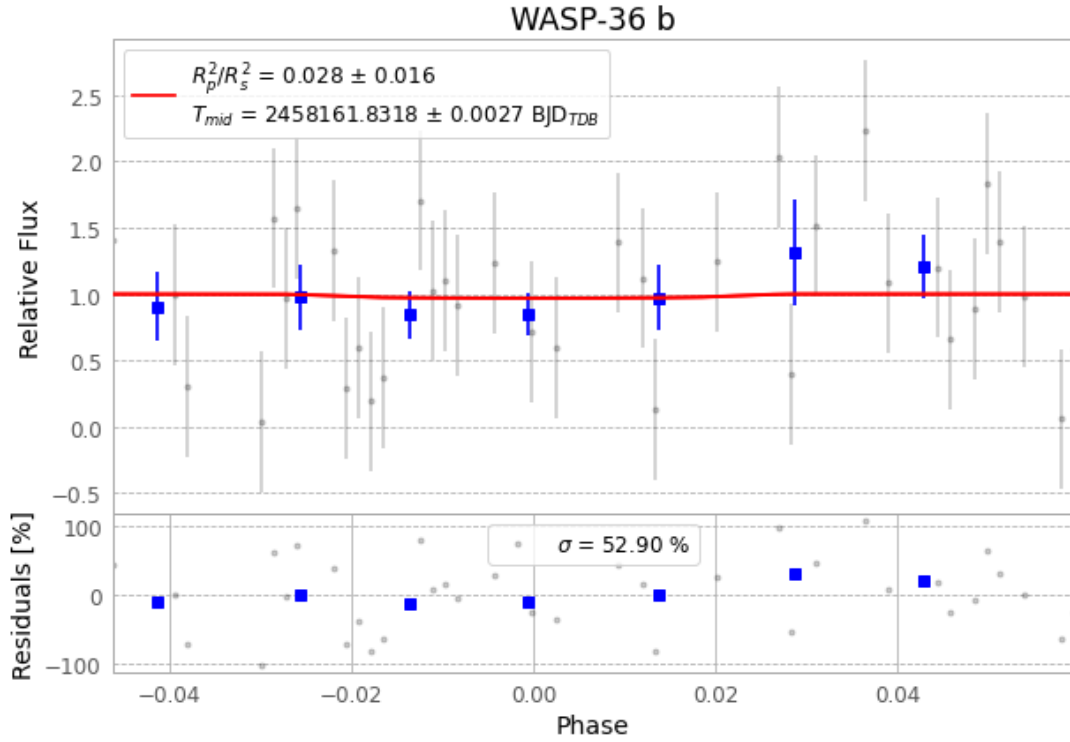


Figure 1 — FinalLightCurve_WASP-36 b_2018-02-11.png (SHA-256 d263d8e3c745cad3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [503, 447], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2627ee6fec9f3f0b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

79 FITS frames (52 MB), WASP-36180212060612.FITS → WASP-36180212100012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-180212.log (3ee5dfa7cddb...), FinalLightCurve_WASP-36 b_2018-02-11.png (d263d8e3c745...), FinalLightCurve_WASP-36 b_2018-02-11.csv (a7c4893de225...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 23:58Z.